

FOS 709 Tutorial Laboratory 2

Exploratory Data Analysis: Common Graphical and Numerical Summary Estimates

NAME:

1. Explore the **linton** data set:

- a. Generate a histogram and box-and-whiskers plot for the Duration of the first Segment of Genuine (GEN) signatures in the linton data set.
- b. Do the same as 1a. for the Disguised (DNM) signatures.
- c. Compute the mean for both (sub) datasets in 1a. and 1b.
- d. Compute the median for both (sub) datasets in 1a. and 1b.
- e. Compute the standard deviation for both (sub) datasets in 1a. and 1b.
- f. Compute the median absolute deviation for both (sub) datasets in 1a. and 1b.
- g. Compute the range for both (sub) datasets in 1a. and 1b.
- h. How do these (sub) datasets compare?

2. Explore the **gas** data set:

- a. Using the **pairs()** function, make a pairs-plot of the first six chemical components of the data set: Ethylbenzene, m.p.Xylene, o.Xylene, Propylbenzene, m.Ethyltoluene.
- b. How would you describe the relationship between these components, if any?
- c. Using the **stack()** function, stack the columns of the first five chemical components of **gas**. Next use the **histogram()** function in the **lattice** package to make a histogram array for the abundances of these five components.
- d. Use the **bwplot()** function in the **lattice** package to make an array of box-and-whiskers plots for the five components.
- e. Which component abundances seem most different?

3. Explore the **shedder** data set:

- a. Compute the average amount of touch DNA shed for each participant at each location/wash status.
- b. Compute the standard deviation of touch DNA shed for each participant at each location/wash status.
- c. Compute the coefficient of variation (as a percentage) for the amount of touch DNA shed by each participant at each location/wash status.
- d. Plot an array of histograms for the coefficient of variations at each location. Comment about the variation of the amount of touch DNA recovered relative to the average at each location.
- e. Compute the range (difference) of amount of touch DNA shed by each participant at each location/wash status. What is the min/max range. Plot a histogram of these ranges. Comment.